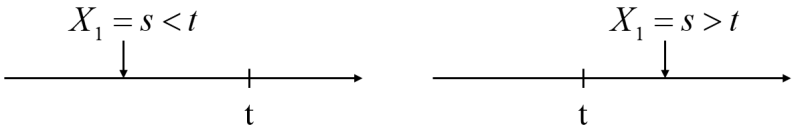
作业 4

第一题



（1）

$$M(t) = E[N(t)] = E[E(N(t)|X_1)]$$

$$= \int_0^{+\infty} E[N(t)|X_1 = s]dF(s)$$

$$= \int_0^t E[N(t)|X_1 = s]dF(s) + \int_t^{+\infty} E[N(t)|X_1 = s]dF(s)$$

$$= \int_0^t E[1 + N(t-s)]dF(s) + \int_t^{+\infty} E[0|X_1 = s]dF(s)$$

$$\therefore M(t) = F(t) + \int_0^t M(t-s)dF(s)$$

（2）代入$F(t)$的表达式

又$\because X_i \sim UNIF(0,1)$

$$\therefore F(t) = \begin{cases} 0, t < 0 \\ t, 0 \le t \le 1 \\ 1, t > 1 \end{cases}$$

$\therefore$当 $0 \le t \le 1$ 时

$$M(t) = t + \int_0^t M(t-s)ds \xrightarrow{令t-s=u} t + \int_0^t M(u)du$$

$$\therefore \begin{cases} M'(t) = 1 + M(t) \\ M(0) = E[N(0)] = 0 \end{cases}$$

第二题

$$P(N(2) = 1) = P(T_1 \le 2 < T_2) = P(X_1 \le 2 < X_1 + X_2)$$

$$= \sum_k P(X_1 = k)\, P(X_1 \le 2 < X_1 + X_2 | X_1 = k)$$

$= \sum_k P(X_1 = k)\, P(k \le 2 < k + X_2 | X_1 = k)$

$= \sum_{k\le 2} P(X_1 = k)\, P(2 - k < X_2) = P(X_1 = 1)P(X_2 > 1) + P(X_1 = 2)P(X_2 > 0)$

$= P(X_1 = 1) \cdot P(X_2 = 2) + P(X_1 = 2) \cdot 1$

结论： $\because X_i \sim \begin{pmatrix} 1 & 2 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$ 脱掉条件概率符号是因为 X_1 和 X_2 是**相互独立**的。

$\therefore P(N(2) = 1) = \frac{1}{3} \cdot \frac{2}{3} + \frac{2}{3} = \frac{8}{9}.$

或者：

做法 2：

$P(N(2) = 1) = F^{(1)}(2) - F^{(2)}(2)$
$= F(2) - F^{(2)}(2)$
$= P(X_1 \le 2) - P(X_1 + X_2 \le 2)$
$= 1 - P(X_1 + X_2 \le 2) = 1 - P(X_1 = 1, X_2 = 1) = 1 - P(X_1 = 1) \cdot P(X_2 = 1)$

结论： $\because X_i \sim \begin{pmatrix} 1 & 2 \\ \frac{1}{3} & \frac{2}{3} \end{pmatrix}$

$\therefore P(N(2) = 1) = 1 - \frac{1}{3} \cdot \frac{1}{3} = \frac{8}{9}.$

同理：(采用想法 2)

$P(N(3) = 2) = F^{(2)}(3) - F^{(3)}(3) = P(X_1 + X_2 \le 3) - P(X_1 + X_2 + X_3 \le 3)$

$= P(X_1 = 1, X_2 = 1) + P(X_1 = 1, X_2 = 2) + P(X_1 = 2, X_2 = 1)$

$\quad -P(X_1 = 1, X_2 = 1, X_3 = 1)$

$\quad = \frac{1}{3} \cdot \frac{1}{3} + \frac{1}{3} \cdot \frac{2}{3} \cdot 2 - \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{14}{27}.$

第三题：

证明：$\forall 0 \le x \le t$，利用“$N(t) = k$”构造完备事件组，简化$P\left(T_{N(t)} \le x\right)$。

$$P\left(T_{N(t)} \le x\right) = \sum_{k=0}^{+\infty} P\left(T_{N(t)} \le x, N(t) = k\right)$$

$= P(N(t) = 0) + \sum_{k=1}^{+\infty} P\left(T_k \le x, N(t) = k\right)$

$$= P(X_1 > t) + \sum_{k=1}^{+\infty} P\left(T_k \le x, T_k \le t < T_{k+1}\right)$$

$$= \bar{F}(t) + \sum_{k=1}^{+\infty} P\left(T_k \le x, T_k + \mathrm{X_{k+1}} > \mathrm{t}\right)$$

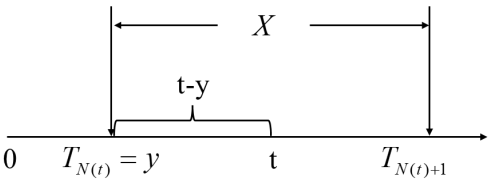
$$= \bar{F}(t) + \sum_{k=1}^{+\infty} \int_0^{+\infty} P\left(T_k \le x, T_k + \mathrm{X_{k+1}} > \mathrm{t} | T_k = \mathrm{u}\right) dF^{(k)}(u)$$

$$= \bar{F}(t) + \sum_{k=1}^{+\infty} \int_0^{x} P\left(X_{k+1} > t - u\right) dF^{(k)}(u)$$

$$= \bar{F}(t) + \int_0^{x} \bar{F}(t-u) d\left[\sum_{k=1}^{+\infty} F^{(k)}(u)\right]$$

$$= \bar{F}(t) + \int_0^{x} \bar{F}(t-u) dM(u)$$

第四题



(a)

$$P\left(N(t) = n \middle| T_n = y\right) = P\left(T_n \le t < T_{n+1} \middle| T_n = y\right)$$

$$= P\left(T_n \le t < T_n + X_{n+1} \middle| T_n = y\right)$$

$$= P\left(y \le t < y + X_{n+1} \middle| T_n = y\right)$$

$$= \begin{cases} 0, & y > t \\ P\left(X_{n+1} > t - y\right), & y \le t \end{cases}$$

$$= \begin{cases} 0, & y > t \\ 1 - F(t-y), & y \le t \end{cases}$$

(b)

$$P(N(t) = n) \xrightarrow{\text{全概率公式}} \int_0^{+\infty} P\left(N(t) = 1 \middle| T_n = y\right) dF^{(n)}(y)$$

$$= \int_0^{t} \left[1 - F(t-y)\right] dF^{(n)}(y) + \int_t^{+\infty} 0 \cdot dF^{(n)}(y)$$

$$= \int_0^{t} \left[1 - F(t-y)\right] dF^{(n)}(y) \qquad (1)$$

$\because X_i \sim Exp(\lambda)$，且$X_1, X_2, \ldots, X_i, \ldots$互独，

$\therefore F(t-y)=1-e^{-\lambda(t-y)}$，$T_n=X_1+X_2+\ldots+X_n\sim Gamma(n,\lambda)$

$\therefore dF_{T_n}(y)=dF^{(n)}(y)=\frac{\lambda(\lambda y)^{n-1}}{(n-1)!}e^{-\lambda y}dy$（Gamma 分布的密度）

将上式代入(1)得：

$$P(N(t)=n)=\int_0^t e^{-\lambda(t-y)}\frac{\lambda(\lambda y)^{n-1}}{(n-1)!}e^{-\lambda y}dy=e^{-\lambda t}\int_0^t\frac{\lambda(\lambda y)^{n-1}}{(n-1)!}dy$$

$=\frac{(\lambda t)^n}{n!}e^{-\lambda t}$.（恰为 Poisson 分布，即$N(t)\sim P(\lambda t)$，此时更新过程退化为 Poisson 分布。

做法 2：

$$P(N(t)=n)=P\left(T_n\le t<T_{n+1}\right)=P\left(T_n\le t<T_n+X_{n+1}\right)$$

$$=\int_0^{+\infty}P\left(T_n\le t<T_n+X_{n+1}\middle|T_n=x\right)dF^{(n)}(x)$$

$$=\int_0^{+\infty}P\left(x\le t<x+X_{n+1}\right)dF^{(n)}(x)$$

$$=\int_0^t P\left(t-x<X_{n+1}\right)dF^{(n)}(x)$$

$$=\int_0^t\left[1-F(t-x)\right]dF^{(n)}(x)$$

余下部分与做法 1 一致。

第五题

$$P(N=n)=P\left(U_1+\ldots+U_{n-1}\le 1<U_1+\ldots+U_n\right)$$

$$=\int_0^1 P\left(U_1+\ldots+U_{n-1}\le 1<U_1+\ldots+U_n\middle|U_1+\ldots+U_{n-1}=x\right)dF^{(n-1)}(1)$$

$$=\int_0^1 P\left(U_n>1-x\right)dF^{(n-1)}(x)$$

$$=\left[1-F(1)\right]*F^{(n-1)}(1)$$

$$=F^{(n-1)}(1)-F^{(n)}(1)$$

$\because P(R(1)=N-1=n)=P(N=n+1)=F^{(n)}(1)-F^{(n+1)}(1)$

$\therefore\{R(t),t\ge 0\}$可以被看作以$\{U_i,i\ge 1\}$为更新间隔序列的更新过程。

记$M(t)=E[R(t)]$，$\because U_i\sim UNIF(0,1)$，故由本次作业第 1 题得：

$M(t)=e^t-1,\ 0\le t\le 1$，即$M(1)=E[N]-1=e-1$

$\therefore E[N]=e.$

或者，关于“$P(N=n)=F^{(n-1)}(1)-F^{(n)}(1)$”，还可以直接计算卷积：

$F^{(n)}(t)\left(F(t)=\begin{cases}0,\ x<0\\ t,0\ \le\ t\ \le\ 1\\ 1,\ t>1\end{cases}\right)$

$F^{(2)}(t)=P(X_1+X_2\le t)$

$=\int_0^t P(X_1+X_2\le t|X_1=x)dF(x)$

$=\int_0^t P(X_2\le t-x)dx=\int_0^t F(t-x)dx=\int_0^t (t-x)dx=\frac{t^2}{2!}.$

$F^{(3)}(t)=P(X_1+X_2+\mathrm{X}_3\le t)$

$=\int_0^t P(X_1+X_2\le t-x|X_3=x)dF(x)$

$=\int_0^t P(X_1+X_2\le t-x)dx=\int_0^t F^{(2)(t-x)}dx=\int_0^t \frac{(t-x)^2}{2}dx=\frac{t^3}{3!}.$

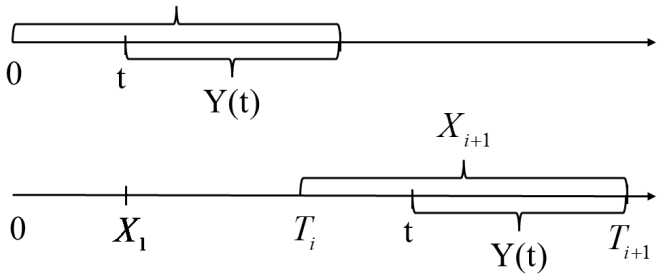
……

$F^{(n)}(t)=\frac{t^n}{n!},0\le t\ \le 1.$

$\therefore P(N=n)=F^{(n-1)}(1)-F^{(n)}(1)=\frac{t^{n-1}}{(n-1)!}-\frac{t^n}{n!}$

$E[N]=\sum_{n\ge 1} n\cdot P(N=n)=e.$

第六题



做法 1：

$K(t) = \int_{-\infty}^{+\infty} P(Y(t) \le x | X_1 = s)\, dF(s)$

$= \int_0^t P(Y(t) \le x | X_1 = s)\, dF(s) + \int_t^{+\infty} P(Y(t) \le x | X_1 = s)\, dF(s)$

$= \int_0^t P(Y(t-s) \le x) dF(s) + \int_t^{+\infty} P(X_1 - t \le x | X_1 = s)\, dF(s)$

$= \int_0^t K(t-s) dF(s) + \int_t^{+\infty} P(s \le t+x)\, dF(s) = K(t) * F(t) + \int_t^{t+x} dF(s)$,

$= K(t) * F(t) + \underbrace{F(t+x) - F(t)}_{h(t)}$.

∴由更新方程解的存在 TH 可知

$K(t) = h(t) + h(t) * M(t) = H(t) + \int_0^t h(t-u) dM(u)$

$= F(t+x) - F(t) + \int_0^t [F(t-u+x) - F(t-u)]\, dM(u)$.

做法 2：

$K(t) = P(Y(t) \le x) = P\left(T_{N(t)+1} - t \le x\right) = \sum_{n=0}^{+\infty} P\left(T_{N(t)+1} - t \le x, N(t) = n\right)$

$= P\left(T_{N(t)+1} - t \le x, N(t) = 0\right) + \sum_{n=1}^{+\infty} P\left(T_{N(t)+1} - t \le x, N(t) = n\right)$,

$= P(X_1 \le t + x, X_1 > t) + \sum_{n=1}^{+\infty} P\left(T_n + X_{n+1} \le x + t, T_n \le t \le T_n + X_{n+1}\right)$

$= F(t+x) - F(t) + \sum_{n=1}^{+\infty} \int_0^{+\infty} P\left(X_{n+1} \le x + t - T_n, T_n \le t \le T_n + X_{n+1} | T_n = u\right) dF^{(n)}(u)$

$= F\left(t+x\right) - F\left(t\right) + \int_0^t P\left(t - u < X_{n+1} \le t + x - u\right) dM\left(u\right)$

$$= F(t+x) - F(t) + \int_0^t F(t+x-u) - F(t-u)\, dM(u).$$